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Experiment-30

Algorithm

1. Start.
2. Input the number of elements n .
3. Input n array elements.
4. Display the menu repeatedly until the user chooses Exit.
5. Read the user's choice.
6. Use `switch-case`:
 - **Choice 1:** Sort the array in ascending order using Bubble Sort. Input the element to search and perform Binary Search.
 - **Choice 2:** Ask whether the user wants insertion or deletion.
 - For insertion, input a valid position and value, shift elements to the right, and insert the value.
 - For deletion, input a valid position, shift elements to the left, and reduce the array size.
 - **Choice 3:** Input k and rotate the array to the right k positions.
 - **Choice 4:** Exit the program.
 - Otherwise, display "Invalid choice".
7. Stop.

Test Case Table

TC	Array	Choice	Additional Input	Expected Output
TC1	50 20 40 10 30	1	Search 40	Sorted: 10 20 30 40 50; 40 found
TC2	10 20 30 40	1	Search 25	Sorted: 10 20 30 40; 25 not found
TC3	10 20 30 40	2	Insert 25 at position 3	10 20 25 30 40
TC4	10 20 30 40 50	2	Delete position 3	10 20 40 50
TC5	10 20 30 40 50	3	$k = 2$	40 50 10 20 30
TC6	10 20 30	2	Invalid position 8	Invalid position
TC7	10 20 30	5	—	Invalid choice

C Program

```
#include <stdio.h>
```

```

int main()
{
    int a[100], n, choice;
    int i, j, temp;
    int key, low, high, mid, found;
    int pos, value;
    int k;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Bubble Sort and Binary Search\n");
        printf("2. Insertion and Deletion\n");
        printf("3. Right Rotation by k positions\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                /* Bubble Sort */
                for(i = 0; i < n - 1; i++)
                {
                    for(j = 0; j < n - i - 1; j++)
                    {
                        if(a[j] > a[j + 1])
                        {
                            temp = a[j];
                            a[j] = a[j + 1];
                            a[j + 1] = temp;
                        }
                    }
                }

                printf("Sorted array: ");
                for(i = 0; i < n; i++)
                {
                    printf("%d ", a[i]);
                }
            }
        }
    }

```

```

/* Binary Search */
printf("\nEnter element to search: ");
scanf("%d", &key);

low = 0;
high = n - 1;
found = 0;

while(low <= high)
{
    mid = (low + high) / 2;

    if(a[mid] == key)
    {
        found = 1;
        break;
    }
    else if(a[mid] < key)
    {
        low = mid + 1;
    }
    else
    {
        high = mid - 1;
    }
}

if(found == 1)
    printf("%d found at position %d\n", key, mid + 1);
else
    printf("%d not found in the array.\n", key);

break;

```

```

case 2:
    printf("\n1. Insertion\n");
    printf("2. Deletion\n");
    printf("Enter your choice: ");
    scanf("%d", &j);

    if(j == 1)
    {
        printf("Enter position for insertion (1 to %d): ", n + 1);
        scanf("%d", &pos);

        if(pos < 1 || pos > n + 1)
        {
            printf("Invalid position.\n");

```

```

    }
else
{
    printf("Enter value to insert: ");
    scanf("%d", &value);

    for(i = n; i >= pos; i--)
    {
        a[i] = a[i - 1];
    }

    a[pos - 1] = value;
    n++;

    printf("Array after insertion: ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    printf("\n");
}
}
else if(j == 2)
{
    printf("Enter position for deletion (1 to %d): ", n);
    scanf("%d", &pos);

    if(pos < 1 || pos > n)
    {
        printf("Invalid position.\n");
    }
    else
    {
        for(i = pos - 1; i < n - 1; i++)
        {
            a[i] = a[i + 1];
        }

        n--;

        printf("Array after deletion: ");
        for(i = 0; i < n; i++)
        {
            printf("%d ", a[i]);
        }
        printf("\n");
    }
}
else
{

```

```

        printf("Invalid choice.\n");
    }

    break;

case 3:
    printf("Enter number of positions to rotate: ");
    scanf("%d", &k);

    k = k % n;

    for(j = 0; j < k; j++)
    {
        temp = a[n - 1];

        for(i = n - 1; i > 0; i--)
        {
            a[i] = a[i - 1];
        }

        a[0] = temp;
    }

    printf("Array after right rotation: ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    printf("\n");

    break;

case 4:
    printf("Program terminated.\n");
    break;

default:
    printf("Invalid choice. Please try again.\n");
}

} while(choice != 4);

return 0;
}

```